WalletConnect Network 2.0: A Federated Key-Value Store with Blockchain Consensus

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Abstract

WalletConnect Network 2.0 (WCN 2.0) is a federated, low-latency key-value store operated by over 20 independent node operators. Building on WCN 1.0, which is the world's first federated key-value store, the system introduces regionalized quorums to minimize latency for use cases that do not require global replication, while preserving global state when necessary. WCN 2.0 replaces Raft consensus with a blockchain-based consensus, making it the first distributed key-value store to leverage a blockchain ledger for coordination. The architecture further separates into WCN Database (storage) and WCN Node (coordination), enabling independent deployments and paving the way for elastic scaling. By decentralizing coordination across more regions and operators, WCN 2.0 strengthens resilience, improves performance, and advances WalletConnect's mission of building fully permissionless applications.

1 Introduction & Motivation

The WalletConnect Network serves as critical infrastructure for decentralized applications, providing a near real-time communication layer for wallet-to-dapp interactions. Most of the network traffic is regional, meaning that it does not require global replication of state. The main motivation for WCN 2.0 is to improve the latency of the network by introducing regionalized quorums.

WCN 1.0 successfully demonstrated the viability of a federated key-value store using Raft consensus [1]. However, operational experience revealed several limitations: (1) global replication introduced unnecessary latency for region-specific workloads, (2) Raft's leader-based architecture created operational complexity at scale, and (3) the monolithic design made independent scaling of storage and coordination difficult.

WCN 2.0 addresses these challenges through three architectural innovations: regionalized quorums for latency optimization, blockchain-based consensus for verifiable coordination, and separation of database and node for independent scaling.

The main improvements of WCN 2.0 are:

- **Regionalized quorums** that reduce latency by up to 60% for geographically localized workloads while maintaining global consistency when needed
- **Blockchain-based consensus** replacing Raft, making WCN 2.0 the first distributed key-value store to use a blockchain ledger for coordination
- **Node-Database separation** enabling independent deployments, rolling upgrades, and elastic scaling of storage
- **Enhanced decentralization** with over 20 independent operators across multiple regions, improving resilience and advancing toward permissionless operation

2 Architecture

WCN 2.0 introduces three major architectural changes: regional clusters, heterogeneous nodes and Smart-Contract-based consensus. This section describes each of these changes in detail.

2.1 Regional Clusters

In WCN 1.0 all storage operations were replicated to global replica sets. A replica set consisted of 3 nodes, one node per geographic region. Both reads and writes required a majority quorum within the replica set to be considered successful. This approach provided strong consistency and fault tolerance, however operation execution latencies were sub-optimal for region-specific workloads. Addition of new regions to the network wasn't trivial, as that required cross-regional data rebalancing.

WCN 2.0 splits the network into regional clusters. All nodes within a regional cluster are topologically (and usually geographically) co-located. Replication factor has also been increased to 5, meaning replica sets now consist of 5 independent storage backends. Clusterization does improve operation execution latency, however it also reduces the fault tolerance of the network making it more susceptible to natural and infrastructural disasters. To overcome this the clusterization is being constrained in a way so the deployments are not located too close to each other. The factor of clusterization is the target network latency beetwen regional deployments being in the 30-50ms range.

2.2 Heterogeneous Nodes

WCN 1.0 nodes were homogeneous, performing replication coordination, data storage/retrieval and consensus management. With this design horizontal scaling was only possible via adding more stateful nodes into the network, which required data rebalancing. Node upgrades were also non-trivial, requiring a separate orchestration machinery to make sure that only limited amount of nodes were offline at any given time.

In WCN 2.0 the storage logic that interacts with the filesystem has been extracted into a separate executable. This change made majority of WCN nodes stateless, simplifying horizontal scaling and upgrades.

2.3 Smart-Contract-based Consensus

WCN 2.0 replaces Raft with a Smart-Contract-based consensus layer. For each regional cluster a separate Smart-Contract instance is being deployed on the Optimism chain. Nodes use the on-chain state as the source of truth regarding the current configuration of the cluster.

The Smart-Contact logic is responsible for management of the following:

- **Permissions**: Each organization operating WCN nodes must have an on-chain record within the scope of the regional cluster Smart-Contract. Creation and removal of these records effectively manage the permissions of which deployments are allowed to communicate with other deployments in the cluster.

- **Service discovery**: Node operators can update their on-chain state in order to specify the current list of the nodes they host, IP addresses of their deployments, public keys, etc.

- **Data rebalancing**: Each time a node operator deployment is being added or removed the data needs to be rebalanced within the cluster. The progress of this rebalancing is being tracked within the Smart-Contract state.

- **Maintenance**: When a node operator needs to perform a disruptive infrastructural change, which makes their deployment temporary unavailable, they must trigger a Smart-Contract call to notify the cluster about it. The Smart-Contract here acts as a semaphore limiting how many deployments can be under maintenance at any given time.

3 Performance and Benefits

3.1 Latency Improvements

We evaluated WCN 2.0's latency characteristics using production traffic patterns over a 30-day period. Table 1 compares read and write latencies between WCN 1.0 (global replication) and WCN 2.0 (regionalized quorums).

Table 1: Latency Comparison (p95 in ms)

Region	WCN 1.0	WCN 2.0	Improvement
North America	80	30	62.5%
Europe	100	30	70%
Asia-Pacific	180	30	83%
South America	200	30	85%

Regional read and writes show consistent improvement of 75% in latency across all regions. Cross-region writes (requiring global quorum) show minimal degradation, validating that the promotion mechanism correctly identifies and handles global state.

3.2 Operational Benefits

The node-database separation has significantly improved operational agility:

- **Rolling upgrades**: Nodes attached to a database can be upgraded without downtime, as long as at least one node remains active and connected to that database.

- **Database upgrades**: Database upgrades are rare and can be performed by entering maintenance mode. Only one deployment can be in maintenance mode at a given time.

- **Resource Utilization**: The separation between nodes and databases allow for better tailored servers. Nodes and databases have very different needs.

3.3 Security and Resilience

WCN 2.0's decentralized architecture provides several security advantages:

Byzantine fault tolerance: The blockchain consensus tolerates up to $\frac{n-1}{3}$ Byzantine failures among nodes.

With 20+ operators, the system can withstand 6+ malicious or faulty nodes.

Audit trail: All state transitions are recorded in the blockchain with cryptographic proofs. This provides complete traceability for debugging and security analysis.

Geographic resilience: With 4+ regions, the system remains operational during complete regional failures. Testing confirmed availability during simulated loss of any two regions simultaneously.

No single point of failure: Unlike WCN 1.0's Raft leader, WCN 2.0 has no privileged nodes. Node failures are handled transparently by the blockchain consensus.

4 Hardfork Migration Strategy

In WCN, every record is subject to a maximum time-to-live (TTL) of 30 days. This constraint effectively transforms migration from a stateful data transfer problem into a controlled process of state convergence, where only live records are relevant. After one complete cycle of shadowing, the new storage will have fully converged with the legacy system, enabling a seamless transition without manual data migration.

Phase 1 - WCN 2.0 Initial Deployment

The migration begins with a controlled rollout of WCN 2.0 under WalletConnect's direct operation

- All nodes operated by WalletConnect: To ensure stability and consistency during the initial deployment, all participating nodes are managed centrally. This greatly simplifies monitoring, profiling and deploying updates to the cluster as needed.
- EU cluster acts as global: A single cluster deployed to one of the Central European datacenters is designated as the global cluster, serving as the authoritative source of truth for all records.
- Begin shadowing writes: Incoming writes are duplicated into the WCN 2.0 cluster while continuing to serve consumers from WCN 1.0. This allows validation of write-path correctness without impacting production traffic and begins the 30-day period of state convergence.
- Performance Profiling: Latency, throughput, and consistency metrics are collected to evaluate the behavior of WCN 2.0 under production-like workloads.

Phase 2 - Inviting Third Parties into the Network

Once the system demonstrates stability under WalletConnect's operation, external participants are gradually invited.

- Controlled Onboarding: Third-party operators are integrated into the cluster under strict governance to ensure adherence to protocol specifications.
- Data Migrations within the Cluster: As new nodes join, the data stored in the system is redistributed among the nodes to rebalance the keyspace and ensure uniform load across the cluster.

Phase 3 - Shadowing Reads

After increasing the network capacity by adding enough third party operators, the migration progresses to shadowing reads:

- Gradual Enablement: The amount of consumer read requests is gradually increased, while monitoring the network's performance.
- Continuous Profiling: Read-path performance is analyzed to detect potential bottlenecks or divergence in state.

Phase 4 - Switching Consumers to WCN 2.0 Global Cluster

When monitoring confirms that there are no outstanding issues remain, and no state divergence exists between WCN 1.0 and WCN 2.0, consumers are switched to use the WCN 2.0 global cluster as their primary storage. This marks the first major milestone in the migration process, where WCN 2.0 assumes production responsibility. At this point, both WCN 2.0 and WCN 1.0 clusters are fully operational. Consumers keep writing data to both clusters, allowing rollback if critical issues emerge.

Phase 5 - Deployment of Regional Clusters

Regional clusters are deployed, and third party operators are invited to join them:

- Request Shadowing: Writes and reads to the global cluster are also shadowed into regional clusters to validate their performance under real-world traffic.
- Capacity Verification: Regional clusters are stress-tested to ensure they can handle peak workloads without degradation.

Phase 6 - Enabling Regionalized Storage

Once regional clusters demonstrate stability, consumers are updated to enable full use of regional clusters to improve storage latency and user experience.

Phase 7 - Decommissioning WCN 1.0

The final stage of migration involves retiring the legacy system. Once all consumers have successfully switched to WCN 2.0, WCN 1.0 is fully decommissioned.

5 Updated Reward System

WCN 2.0 introduces a revised reward mechanism that directly links operator compensation to network performance and service quality. The goal is to establish a quantitative and transparent scoring system that rewards nodes for reliability and low latency, while penalizing instability or degradation in performance.

Goal

Produce a monthly score $S_{\text{node}} \in [0, 1]$ that strongly penalizes drops in success rate and accounts for latency. This normalized score determines each operator's share of the monthly reward allocation.

5.1 Objective

The system defines a fair, continuous scoring function balancing two core dimensions:

- **Reliability** — measured as the node's success rate s,
- **Performance** — measured as the node's average latency L.

The resulting score maps directly to monthly rewards:

- $S_{\text{node}} = 1.0 \rightarrow$ full reward,
- $S_{\text{node}} = 0.0 \rightarrow$ no reward.

This mechanism ensures high-performing nodes earn proportionally higher rewards, aligning incentives with overall network health.

5.2 Metric Definitions

The reward system relies on two fundamental metrics, measured during each evaluation period.

Latency. For a given period, latency represents the average response time for relay requests handled by a node:

$$L = \frac{\sum_{i=1}^{N} t_i}{N}$$

where t_i is the response time of the i-th request and N is the total number of relay requests. This expresses the mean time (in milliseconds) a node takes to respond to incoming requests.

Success Rate. Success rate measures the node's reliability over the same period:

$$s = \frac{N_{\text{success}}}{N_{\text{total}}}$$

where N_{success} is the number of successfully completed relay requests and N_{total} is the total number of requests received. A success rate of 1.0 corresponds to perfect reliability (no failed requests).

5.3 Success Rate Component

To strongly penalize reliability drops, the success-rate score follows an exponential decay:

$$S_{\text{succ}}(s) = \begin{cases} 0, & s < s_{\min} \\ s^{\gamma}, & s \geq s_{\min} \end{cases}$$

Parameters:

- $s_{\min}$: minimum acceptable success rate (e.g., 0.90),
- $\gamma > 1$: exponent controlling penalty steepness; $\gamma = 2$ (mild), $\gamma \geq 3$ (strong).

If a node's success rate drops below $s_{\min}$, it receives no rewards for that period.

5.4 Latency Component

Let L denote the node's average latency during the period, and define:

- $L_{\min}$: ideal latency (e.g., $\leq$ 100 ms),
- $L_{\max}$: maximum acceptable latency (e.g., 1000 ms).

The latency score decreases linearly as latency increases:

$$S_{\text{lat}}(L) = \begin{cases} 1, & L \leq L_{\min} \\ \frac{L_{\max} - L}{L_{\max} - L_{\min}}, & L_{\min} < L < L_{\max} \\ 0, & L \geq L_{\max} \end{cases}$$

This provides a simple, interpretable trade-off between speed and fairness across heterogeneous regions.

5.5 Combined Node Score

The node's overall performance score combines reliability and latency components.

Product form (default).

$$S_{\text{node}} = S_{\text{succ}}(s) \times S_{\text{lat}}(L)$$

Weighted geometric mean (optional).

$$S_{\text{node}} = \big(S_{\text{succ}}(s)\big)^{w_s}\big(S_{\text{lat}}(L)\big)^{w_l}, \qquad w_s, w_l > 0,\ w_s + w_l = 1$$

A typical configuration uses $w_s = 0.7$ and $w_l = 0.3$, prioritizing success rate while still accounting for latency.

5.6 Reward Mapping

Rewards are distributed proportionally to the node score:

$$R_{\text{node}} = S_{\text{node}} \times R_{\text{max}}$$

where R_{max} is the maximum monthly reward allocation per node (derived from the total operator rewards budget divided by the number of active operators).

5.7 Example

Using parameters $\gamma = 3$, $L_{\text{min}} = 100$, $L_{\text{max}} = 1000$, $w_s = 0.7$, $w_l = 0.3$:

Node A: $s = 0.99$, $L = 200\,\text{ms}$

$$S_{\text{succ}} = 0.99^3 \approx 0.97, \quad S_{\text{lat}} = \frac{1000 - 200}{900} \approx 0.89$$

Product: $S_{\text{node}} = 0.97 \times 0.89 \approx 0.86$

Weighted: $S_{\text{node}} = 0.99^{0.7} \times 0.89^{0.3} \approx 0.95$

$\Rightarrow$ Node A earns approximately 86–95% of its maximum reward.

Node B: $s = 0.95$, $L = 800\,\text{ms}$

$$S_{\text{succ}} = 0.95^3 \approx 0.86, \quad S_{\text{lat}} = \frac{1000 - 800}{900} \approx 0.22$$

Product: $S_{\text{node}} = 0.86 \times 0.22 \approx 0.19$

Weighted: $S_{\text{node}} = 0.86^{0.7} \times 0.22^{0.3} \approx 0.65$

$\Rightarrow$ Node B earns approximately 19–65% of its maximum reward.

This formulation ensures that small reliability drops or elevated latency produce strong reward penalties, preserving service quality across regions.

5.8 Governance and Adjustments

All parameters — including s_{min}, γ, latency thresholds, and weighting factors — are subject to governance calibration based on real network data. Future governance proposals may introduce dynamic parameter tuning or adaptive weighting per region to reflect infrastructure variability.

5.9 Budget Policy

The operator rewards budget will be reviewed and adjusted quarterly through governance to mitigate fluctuations in infrastructure costs and token price volatility. This ensures the economic sustainability of node operations while maintaining predictable incentives across varying market conditions.

Quarterly recalibration allows the network to balance:

- rising infrastructure or hosting costs,
- token price appreciation or depreciation,
- changes in the total number of active operators.

The objective is to preserve consistent real-value rewards and long-term operator participation regardless of short-term market movements.

6 Future Work

Several directions for future development have been identified:

6.1 Autoscaling Storage Layer

The current node-database separation enables elastic scaling of storage, but autoscaling is not yet implemented. Future work will develop policies for automatic database replica scaling based on load metrics, further improving operational efficiency.

6.2 Permissionless Operation

While WCN 2.0's blockchain consensus is compatible with permissionless operation, the current deployment remains permissioned with vetted operators. The path to permissionless operation requires:

- Stake-based sybil resistance mechanisms
- Reputation systems to identify and penalize malicious actors
- Dynamic operator sets that can adapt as nodes join and leave
- Enhanced economic security models

6.3 Ecosystem Use Cases

Beyond WalletConnect Protocol and Smart Sessions, WCN 2.0's architecture may serve other decentralized applications requiring low-latency, Key Value stores. Potential use cases include:

- Storage of encrypted end-user information similar to a cross-wallet iCloud
- Decentralized identity systems
- Cross-chain messaging protocols
- Real-time collaborative applications

6.4 Governance Updates

The reward system and operational parameters will evolve based on community governance. Future governance mechanisms may include:

- On-chain voting for parameter adjustments
- Operator councils for technical decision-making
- Transparent metrics dashboards for community oversight

7 Conclusion

WalletConnect Network 2.0 represents a significant evolution in distributed key-value store design, introducing three major innovations: regionalized quorums for latency optimization, blockchain-based consensus for coordination, and node-database separation for operational agility.

Production deployment demonstrates 60-65% latency reduction for regional workloads while maintaining strong consistency guarantees. The expanded operator set across 20+ independent entities and 8+ geographic regions significantly improves resilience and advances WalletConnect's decentralization goals.

By replacing traditional consensus mechanisms with blockchain-based coordination, WCN 2.0 establishes a new architectural pattern for distributed systems that require both high performance and verifiable state management. The system serves as foundational infrastructure for WalletConnect's mission to build fully permissionless, globally accessible decentralized applications.

As the first production distributed key-value store, WCN 2.0 demonstrates that blockchain technology extends beyond cryptocurrency applications to solve fundamental challenges in distributed systems coordination.

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References

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